1. Numerical Solutions of Equations

1.1. Approximation

- To find root of a graph, find point where graph passes through x-axis $\therefore$ look for a sign change
- Carry out decimal search
  - Substitute values between where a sign change has occurred
  - Closer to zero, greater accuracy

1.2. Iteration

- To solve equation $f(x)=0$, you can rearrange $f(x)$ into a form $x=\ldots$
- This function represents a sequence that starts at x_0, moving to x_r
- Substitute a value for x_0 and put back into function getting x_1 and so on.
- As you increase r, value becomes more accurate
- Sometimes iteration don't work, these functions are called divergent, and you must rearrange the formula for x in another way.
- For a successful iterative function, you need a convergent sequence.
- Ensure to use the full value and not the rounded off value when carrying out the iteration.

{M16-P32} Question 3:

The equation $x^5-3x^3+x^2-4=0$ has one positive root.

(i) Verify by calculation that this root lies between 1 and 2.

(ii) Show that the equation can be rearranged in the form

$x=\sqrt[3]{\left(3x+\frac{4}{x^2}-1\right)}$

(iii) Use an iterative formula based on this rearrangement to determine the positive root correct to 2 decimal places. Give the result of each iteration to 4 decimal places.

Solution:

Part (i)

Show a sign change and state it:

$(1)^5-3(1)^3+(1)^2-4=-5$

$(2)^5-3(2)^3+(2)^2-4=8$

There is a sign change between the results obtained when the values 1 and 2 are substituted into the equation, therefore the root lies between the values 1 and 2.

Part (ii)

Rearrange the equation:

$$x=\sqrt[3]{\left(3x+\frac{4}{x^2}-1\right)}$$

$$x^3=3x+\frac{4}{x^2}-1$$

$$x^5=3x^3+4-x^2$$

$$x^5-3x^3+x^2-4=0$$

Part (iii)

Carry out the iteration using either one of the values that the root lies in between as the starting point:

$$x_{n+1}=\sqrt[3]{\left(3x_n+\frac{4}{{x_n}^2}-1\right)}$$

$$x_0=1$$

$$x_1=\sqrt[3]{\left(3x_0+\frac{4}{{x_0}^2}-1\right)}=1.8171$$

$$x_2=\sqrt[3]{\left(3x_1+\frac{4}{{x_1}^2}-1\right)}=1.7824$$

$$x_3=\sqrt[3]{\left(3x_2+\frac{4}{{x_2}^2}-1\right)}=1.7765$$

$$x_4=\sqrt[3]{\left(3x_3+\frac{4}{{x_3}^2}-1\right)}=1.7755$$

The positive root = **1.78**

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